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UNIVERSITI TEKNOLOGI MARA

MINI PROJECT ELE526

APPLIED ELECTRONICS CIRCUIT

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1. ABSTRACT

This mini project focuses on the design and simulation of two analog circuit stages to condition low-amplitude sinusoidal signals. The objective is to amplify a 10 mV, 1 kHz sine wave and remove low-frequency noise to produce a clean, usable output. The first part of the project involves a 3-op-amp instrumentation amplifier combined with a precision rectifier to boost and convert the weak AC signal into a rectified DC form. The second part implements a 5th-order Butterworth high-pass filter with a 1 MHz cutoff frequency to eliminate unwanted low-frequency components. Both stages were designed and tested using LTspice. Simulation results show that the amplifier stage successfully achieved a gain of around 100 and accurate signal rectification, while the Butterworth filter effectively removed low-frequency interference. The RH1028 op-amp provided better waveform clarity and cutoff performance compared to the LM741, making the system suitable for applications requiring precise analog signal conditioning.

2. INTRODUCTION

2.1 Background of the Report

Operational amplifiers are essential components in analog electronic design due to their versatility and wide range of applications. They perform signal processing tasks such as amplification, filtering, integration, differentiation, and rectification, making them commonly used in devices like audio amplifiers, sensor interfaces, medical instrumentation, and communication systems. Op-amp configurations vary depending on the application, including voltage followers, comparators, integrators, Schmitt triggers, and instrumentation amplifiers.

Instrumentation amplifiers are particularly useful for dealing with low-level analog signals, particularly those from sensors or transducers. They provide high input impedance, stable gain, and excellent common-mode noise rejection, making them suitable for accurate and noise-resistant signal amplification. Precision rectifiers, which incorporate op-amps, enable accurate rectification even at very small signal amplitudes, making them ideal for low-voltage analog front-end systems. Filtering is necessary when the signal may be affected by low-frequency noise or interference.

High-pass filters, such as the Butterworth high-pass filter, can remove unwanted components while preserving important parts of the signal. A higher-order design, such as a 5th-order Butterworth filter, offers a steeper roll-off, making it more effective at attenuating low-frequency noise without distorting the desired signal. This project focuses on designing and simulating a two-stage analog signal processing system. The first stage amplifies and rectifies a weak 10 mV, 1 kHz sinusoidal signal using an instrumentation amplifier and precision rectifier, while the second stage uses a 5th-order Butterworth high-pass filter to remove low-frequency noise, resulting in a clean output signal.

2.2 Objectives

1. To design an analog signal conditioning system that includes an instrumentation amplifier, a precision rectifier, and a 5th-order Butterworth high-pass filter based on operational amplifier fundamentals.

2. To simulate the designed circuit using LTspice and observe the waveform behavior for each stage.

3. To verify the performance of each circuit block through waveform plots and frequency response analysis, ensuring accurate gain, rectification, and filter characteristics.

4. To evaluate and discuss the simulation results, identify potential issues or limitations, and suggest improvements using standard component values.

2.3 Problem Statements

Low-amplitude analog signals, like 10 mV sine wave at 1 kHz, are often too weak to be processed directly by measurement or conversion systems. Converting these signals to DC is challenging using conventional diode-based rectifiers due to significant signal loss. Low-frequency noise or interference is common in audio and sensor-based systems, making signal clarity crucial. A high-pass filter is commonly used to eliminate unwanted low-frequency components, with the Butterworth filter being preferred for its flat frequency response in the passband. A higher-order design, like a 5th-order Butterworth high-pass filter, offers sharper attenuation beyond the cutoff frequency, removing rumble or drift in the signal. To address these

issues, a complete analog signal conditioning chain is proposed, consisting of a 3-op-amp instrumentation amplifier, a precision rectifier, and a 5th-order Butterworth high-pass filter with a 1 MHz cutoff frequency. The circuit is designed and analyzed using LTspice simulation software to verify gain performance, rectification accuracy, and filtering effectiveness.

2.4 Literature Review

An instrumentation amplifier is specially designed to amplify small differential signals accurately, especially when dealing with weak sensor outputs. The most common setup uses three operational amplifiers, which provide high input impedance, strong resistance to noise because of the common-mode rejection, and easy control of the gain level [1][2]. These features make it suitable for precise signal processing in medical instruments, data acquisition systems, and industrial monitoring where signal accuracy matters [4].

After the signal is amplified, it often needs to be converted from AC to DC before being used by digital circuits or display devices. Standard diode-based rectifiers are not ideal for this purpose, especially when working with low-voltage signals, because of the voltage drop across the diodes. This causes a loss in signal strength. A precision rectifier, which uses op-amps along with diodes, can overcome this limitation. It can handle very small input signals and still perform accurate rectification without loss [3][5]. These circuits are useful in systems where exact signal conversion is necessary for monitoring or control.

To eliminate low-frequency noise or unwanted interference, a high-pass filter is used. Among the many filter types, the Butterworth filter is known for its flat frequency response in the passband, which means it preserves the original signal's amplitude without distortion. For better noise rejection, a 5th-order Butterworth high-pass filter is used in this project, providing a sharper cutoff and better separation between useful signals and low-frequency noise [1][6].

By combining an instrumentation amplifier, a precision rectifier, and a Butterworth high-pass filter, a complete analog front-end circuit can be created to process weak sinusoidal signals accurately before sending them to a display or digital system.

3. METHODOLOGY: INSTRUMENTATION AMPLIFIER + RECTIFIER

3.1 Fundamental Concepts

According to the study, Instrumentation Amplifier is a differential op-amp circuit that provides high input impedance with the convenience of gain adjustment through the variation of a single resistor.

Instrumentation Amplifier with fixed gain :

1. The gain is normally varied by Rg.
2. If the gain variation is not desired, then Rg can be removed, and the differential amplifier can be made with two unity-gain voltage followers by making Rg = ∞

An instrumentation amplifier is an operational amplifier configuration with high capability in amplifying minimal differential signals, usually in the millivolt range. It consists of three op-amps, two as buffers and one as a differential amplifier that allows signal amplification with high stability and accuracy. The main features of this amplifier are very high input impedance, which prevents the influence of the load on the original signal, as well as excellent common-mode rejection (CMRR). Due to these properties, the instrumentation amplifier is very suitable in medical applications (such as ECG), measurement systems, and industrial sensing systems, where accuracy and sensitivity to small signals are crucial. The main advantage of an instrumentation amplifier is the flexibility in setting the amplification value (gain) by simply changing a single resistor, usually known as:

$$\text{Gain} = \left(1 + \frac{2R}{Rg}\right)\left(\frac{Rf}{R1}\right)$$

Figure 3.1 below shows the configuration of an Instrumentation Amplifier, which makes the design simpler and more stable than a typical differential amplifier. Furthermore, it is capable of filtering out external interference such as electrical noise and low-frequency interference, which are often present in electronic systems. In this project, the instrumentation amplifier is designed to amplify a 10 mV sinusoidal signal to a specified voltage with a gain = 100, making it suitable

for subsequent processing by rectifier circuits that require higher voltage levels to function effectively.

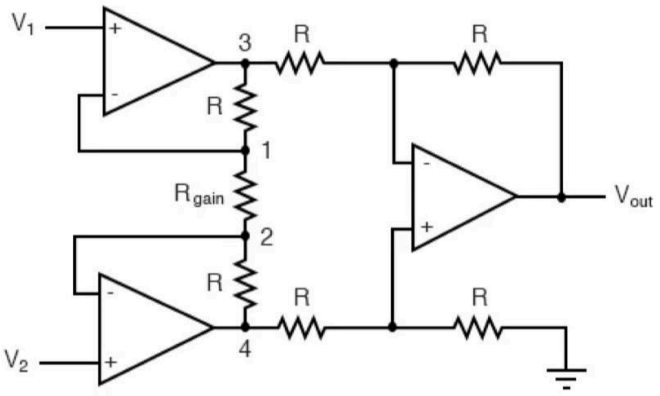


Figure 3.1.1: The configuration of an Instrumentation Amplifier

Referring to the study, a precision rectifier is a circuit that converts alternating current (AC) to direct current (DC). In a typical rectifier circuit, we use diodes to rectify AC to DC. The precision rectifier also converts AC to DC, but in a precision rectifier, an op-amp is used to compensate for the voltage drop across the diode. Precision rectifier or also known as "super diode," is a rectifier circuit that uses an operational amplifier (op-amp) together with a diode to direct a very small AC signal, including in the millivolt range, without being limited by the normal diode threshold voltage (0.7V). In a conventional rectifier, the low-voltage signal cannot be perfectly rectified due to the voltage drop across the diode. Precision rectifiers overcome this problem by using an op-amp to "nudge" that voltage drop, allowing signal rectification with voltages as low as a few millivolts. It can be designed as a half-wave or full-wave rectifier, depending on the application requirements. The configuration of a half-wave and full-wave rectifier is shown in Figures 3.1 and 3.2 below.

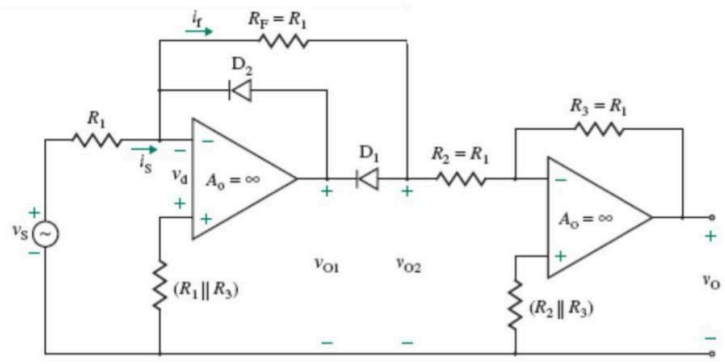


Figure 3.1.2: The configuration of a half-wave rectifier

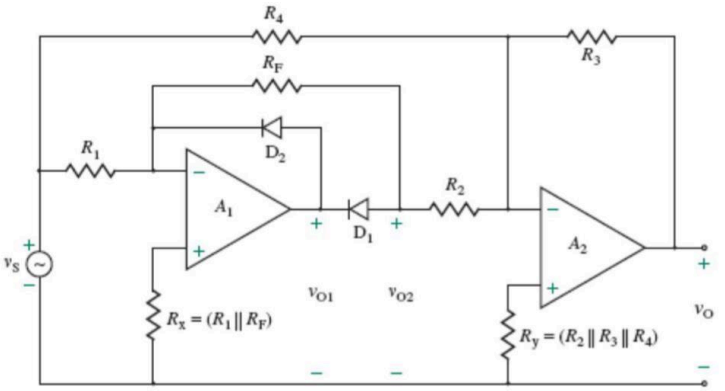


Figure 3.1.3: The configuration of a full-wave rectifier

In electronic systems that require measurement and processing of small AC signals, such as audio detectors, sensors, and communication systems, precision rectifiers are very useful for producing DC or half-DC outputs that represent the true amplitude of the original signal. In this project, a precision rectifier is used to rectify the signal that has been amplified by an instrumentation amplifier. Since the original signal is only 10 mV, the use of a precision rectifier is important to ensure that the rectification process occurs accurately, without losing voltage information due to the threshold voltage of the rectifier components. In addition, this design ensures high accuracy in measurement and control applications.

3.2 Theoretical Calculations

- Gain Calculation (100x for 10 mV input)

In this project, the input signal used is a sinusoidal wave with an amplitude of 10 mV (0.01 V) and a frequency of 1 kHz. The main purpose of this design is to amplify this small signal 100 times using an instrumentation amplifier.

Instrumentation Amplifier

Op-amp golden rules;

$V_d = V_p - V_n = 0V$

$i_1 = 0A$

Analysis current i_{rg};

$$i_{rg} = \frac{V_{o1} - V_{o2}}{2R + R_g}$$

$$i_{rg} = \frac{V_{s1} - V_{s2}}{R_g}$$

$$\frac{V_{s1} - V_{s2}}{R_g} = \frac{V_{o1} - V_{o2}}{2R + R_g}$$

$$\frac{V_{s1} - V_{s2}}{R_g} = \frac{V_{o1} - V_{o2}}{2R + R_g}$$

$$V_{o1} - V_{o2} = (V_{s1} - V_{s2}) \frac{2R}{2R + R_g}$$

$$V_{o1} - V_{o2} = (V_{s1} - V_{s2}) \left(1 + \frac{2R}{R_g}\right)$$

$$V_o = V_{o2} - V_{o1} \left(\frac{R_F}{R_1}\right)$$

$$V_o = -(V_{s1} - V_{s2}) \underbrace{\left(1 + \frac{2R}{R_g}\right)\left(\frac{R_F}{R_1}\right)}_{\text{Gain}}$$

Given gain 100,

$$100 = \left(1 + \frac{2R}{R_g}\right)\left(\frac{R_F}{R_1}\right)$$

$$100 = \left(1 + \frac{2(10k)}{R_g}\right)\left(\frac{10k}{10k}\right)$$

$R = 10\ k\Omega$

$$\left(\frac{2(10k)}{R_g} + 1\right)\left(\frac{100k}{10k}\right) = 100$$

$$\left(\frac{20k}{R_g} + 1\right) 10 = 100$$

$$\frac{20k}{R_g} + 1 = 10$$

$$\frac{20k}{R_g} = 9$$

$$R_g = 2.22k$$

$$V_o = -(V_{s1} - V_{s2}) \left(1 + \frac{2R}{R_g}\right)\left(\frac{R_F}{R_1}\right)$$

$$= -(10m - 20m)\left(1 + \frac{2(10k)}{2.22k}\right)\left(\frac{100k}{10k}\right)$$

$$= 1.009\ V$$

By adding a precision half or full rectifier after the instrumentation amplifier with Vout = 1V, the expected output waveform is as

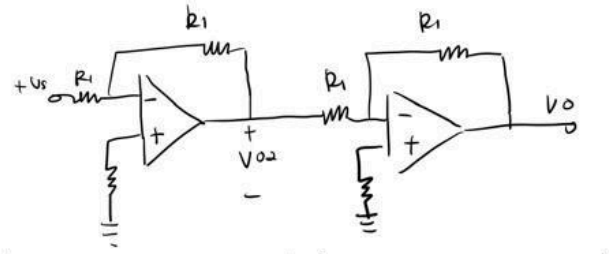
If the precision half wave rectifier,

Consider positive cycle

$V_s = +V_s$

D_2 OFF

D_1 ON



Inverting amp. | Inverting amp.

$V_{o2} = -\frac{R_1}{R_1} \cdot +V_s = -V_s$

$= -\frac{10k}{10k} \cdot 1V = -1V$

$V_o = -\frac{R_1}{R_1} \cdot (-V_s) = +V_s$

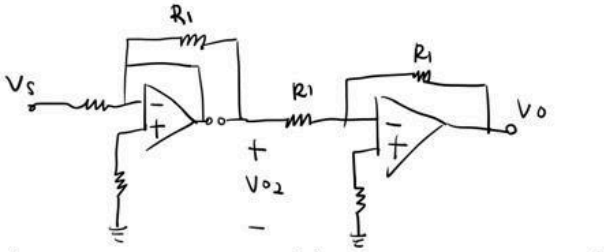
$= -\frac{10k}{10k} \cdot (-1V) = 1V$

Consider positive cycle

$V_s = +V_s$

D_2 ON

D_1 OFF

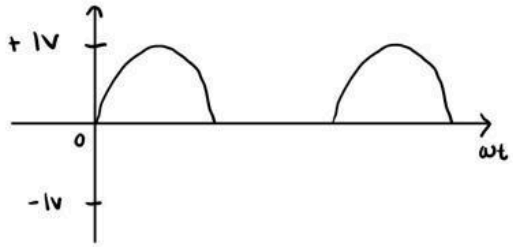


Incomplete amp. | If no volt supply

So $V_{o2} = 0V$

So $V_o = 0V$

Expected Output Waveform For Half Wave Rectifier



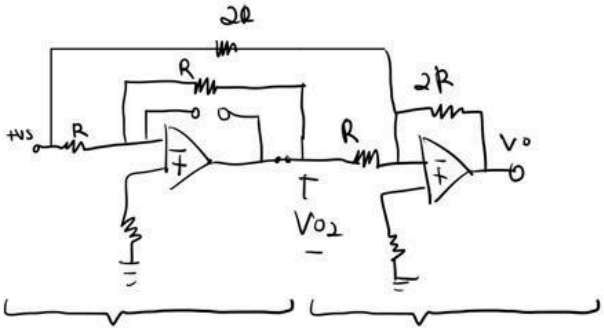
If the precision full wave rectifier,

Consider positive cycle

$V_S = +V_S$

D_2 OFF

D_1 ON



Inverting amp

Summing amp with two input.

$V_{o2} = -\frac{R_1}{R_1} \cdot V_S = -V_S$

$= -\frac{10k}{10k} \cdot 1V = -1V$

$V_o = \frac{-2R}{2R} \cdot V_S - \frac{2R}{R} \cdot (-V_S)$

$= -V_S + 2V_S = +V_S$

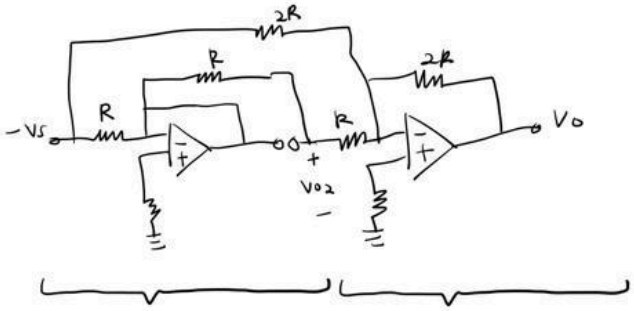
$= -1V + 2(1V) = +1V$

Consider positive cycle

$V_S = +V_S$

D_2 ON

D_1 OFF



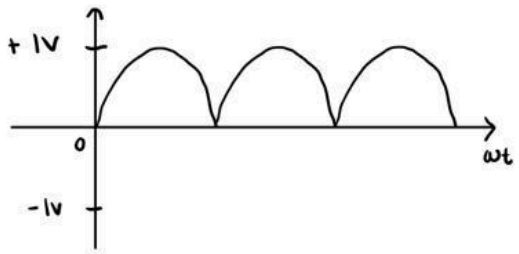
Incomplete amp

So $V_{o2} = 0V$

Summing amp

$V_o = \frac{-2R}{2R}(-V_S) - \frac{2R}{R}(0) = +V_S$

Expected Output Waveform For Full Wave Rectifier



3.3 Circuit Design and Assumptions

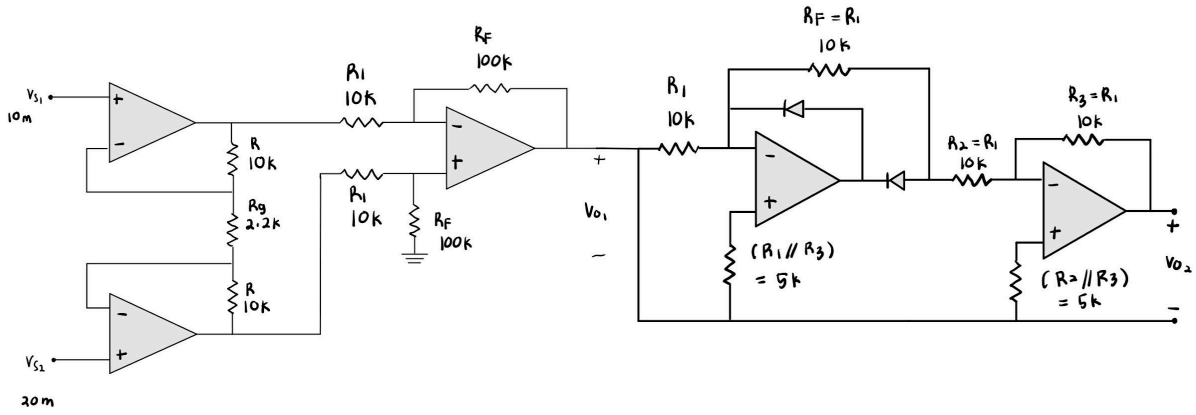


Figure 3.3.1: Instrumentation amplifier with half-wave rectifier design assumption

Figure 3.3.1 shows an Instrumentation amplifier with a half-wave precision rectifier circuit, which contains two main parts, the instrumentation amplifier and the precision rectifier. In the amplifier part, the first two op-amps function as buffers, while the third op-amp (in the middle) functions as a differential amplifier. The resistor value R = 10 kΩ is used consistently to ensure balance and easy, also accurate gain calculation. The value Rgain = 2.2 kΩ is chosen to obtain an amplification of almost 100 times. In the rectifier section, the resistors R1, R2, R3, and Rf are each arranged so that the values of Rx = R1 || R3 and Ry = R2 || R3 are 5k Ω, which helps ensure output stability and makes it easier for the designer to control the waveform generated.

The use of the LM741 op-amp is based on its suitability for basic analog applications that do not require high speed. Although the LM741 is not the fastest op-amp, it is sufficient for low frequencies such as 1 kHz in this project. In addition, the LM741 is readily available and stable enough to be used in this type of rectifier circuit. The assumption that all resistors have a value of 10kΩ (except Rgain) simplifies the mathematical analysis and ensures a high impedance input and a clean output without too much noise or waveform distortion.

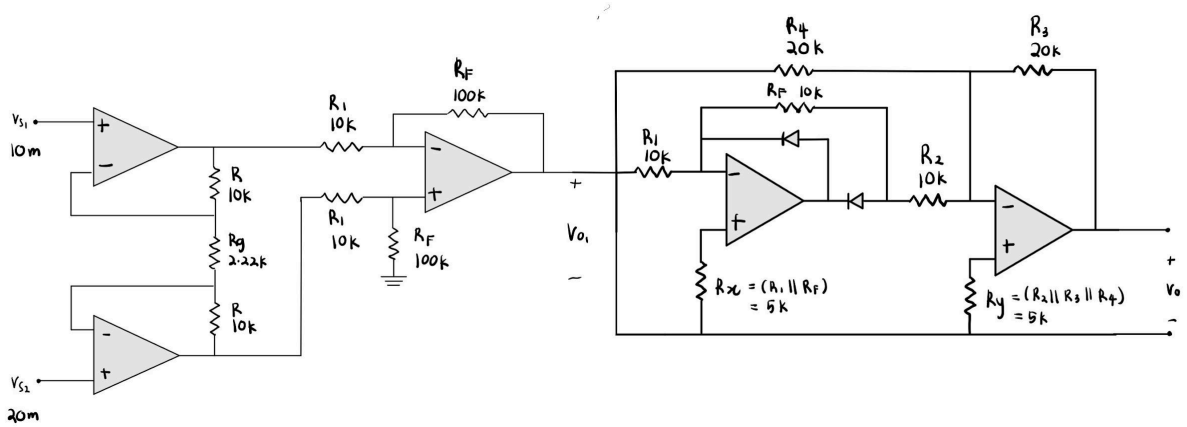


Figure 3.3.2: Instrumentation amplifier with full-wave rectifier design assumption

Figure 3.3.2 shows an instrumentation amplifier with a full-wave rectifier circuit. This configuration is more complex than a half-wave because it involves two stages of rectification. The first part remains the same, namely the instrumentation amplifier with Rgain = 2.2 kΩ, determined by the given gain of 100 and a fixed resistor R = 10 kΩ. The rectifier part uses two additional op-amps specially arranged with four resistors (R1, R2, R3, R4) and two diodes, to ensure that both the positive and negative parts of the signal can be rectified into a full waveform. The resistor values are also assumed to be the same (10kΩ and 20kΩ) so that the voltage is divided symmetrically, thus ensuring a stable and accurate output shape.

In this design, the selection of the LM741 is appropriate because the frequency of 1 kHz is within the linear capability range of this op-amp. Although not the best choice for high-frequency applications, the LM741 is still sufficient to process a simple sine waveform without experiencing large slewing or distortion. The assumption of using balanced resistor values (R1 = R2 = 10 kΩ and R3 = R4 = 20 kΩ) also ensures circuit stability and facilitates analysis in calculating gain, voltage division, and achieving full waveform accuracy in the output.

4. RESULTS AND DISCUSSION: INSTRUMENTATION AMPLIFIER + RECTIFIER

This section presents the results and discussion of an instrumentation amplifier circuit combined with a precision rectifier, designed and simulated using LTSpice. The system is designed to amplify a small difference input signal by 100 times, followed by a rectifier that converts the sine signal to a partial or full positive form. The instrumentation amplifier is built with three op-amps in a standard configuration and uses an ideal op-amp model to obtain a response that is independent of the practical limitations of the actual components.

The simulation results in figures 4.1 and 4.2 below show that the instrumentation amplifier successfully boosts a 10mV sine signal to around 1V, in line with expectations based on theoretical calculations. This signal is then fed to a precision rectifier using an op-amp configuration to rectify the signal without the forward voltage drop that is common with conventional diodes. The simulation results show that the rectifier produces an almost perfect waveform, demonstrating rectification efficiency at low voltage levels. Overall, the behavior of the system is consistent with theory, with only minor variations that may be due to the limitations of the ideal simulation modeling.

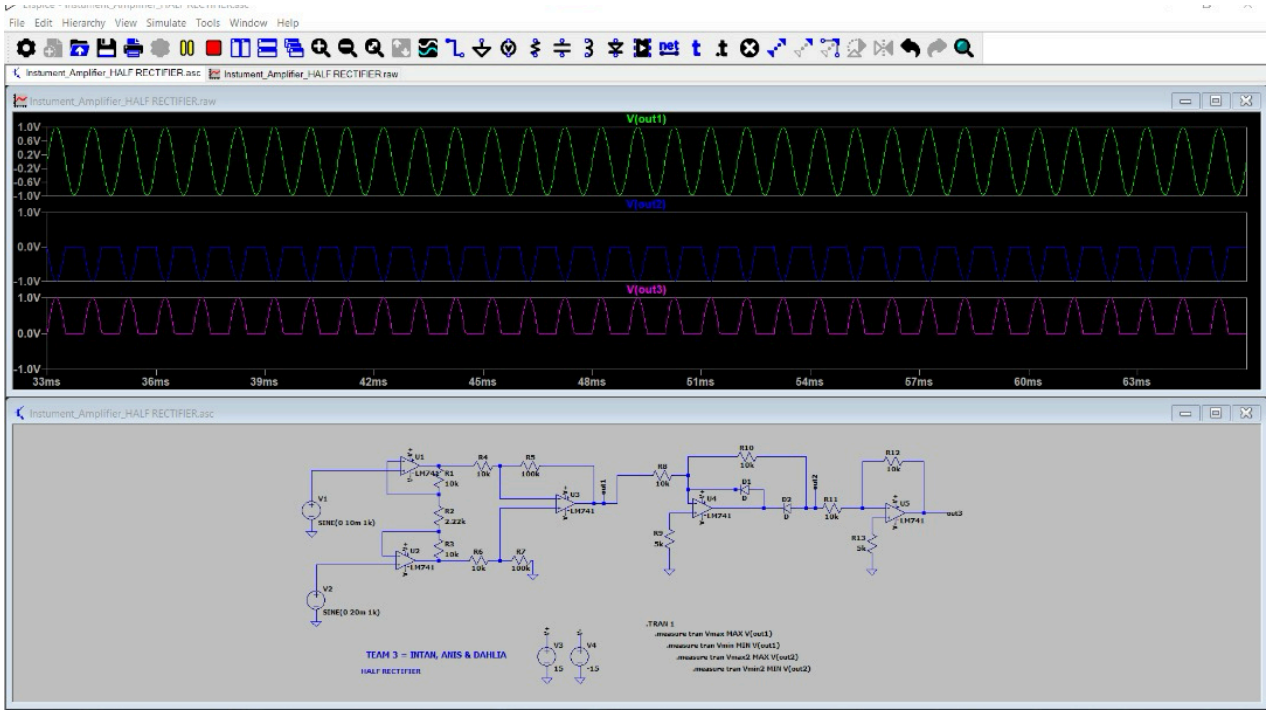


Figure 4.1: Output waveform and circuit for instrument amplifier with half-wave rectifier

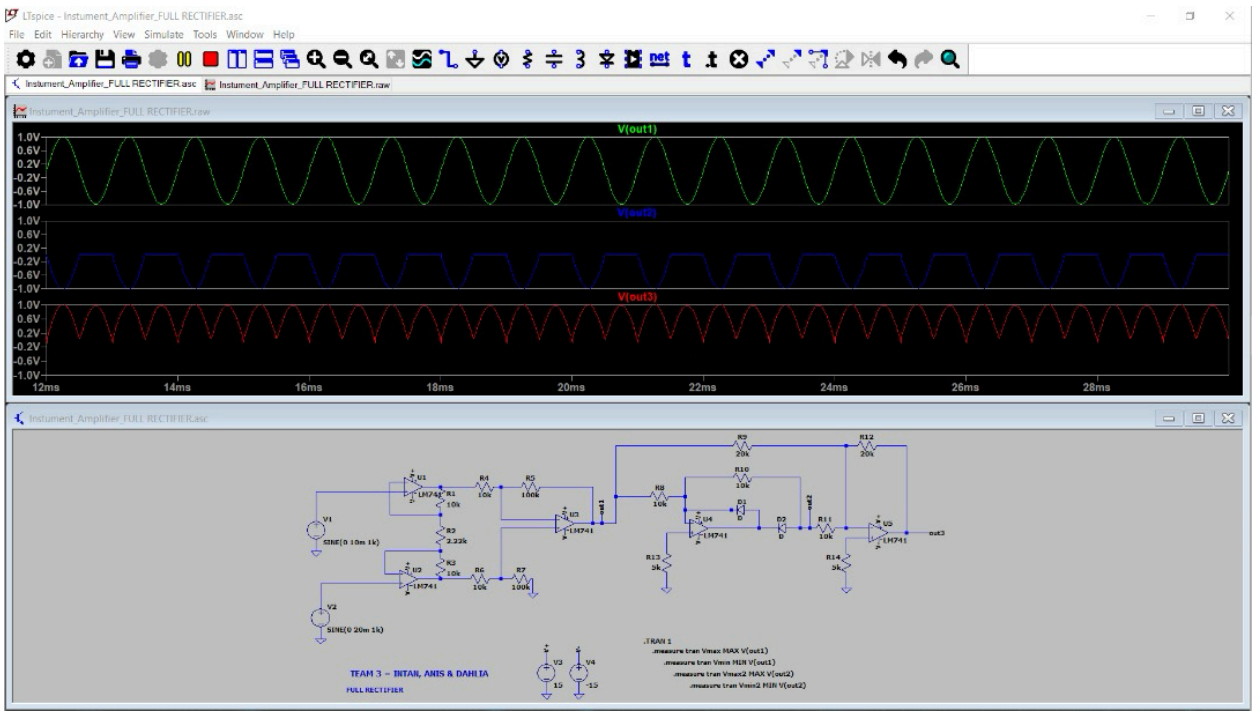


Figure 4.2: Output waveform and circuit for instrument amplifier with half-wave rectifier

Figure 4.1 and Figure 4.2 show the instrumentation amplifier circuit connected to a half-wave rectifier and a full-wave rectifier. Three main output points are observed, namely Vout1 (after the instrumentation amplifier), Vout2 (after the rectifier diode), and Vout3 (after the second op-amp). Vout1 shows the original signal that has been amplified symmetrically, including both the positive and negative parts of the wave. This point is important to use as a reference to evaluate how the shape of the original signal changes after going through the rectification process.

At point Vout2, the processed signal comes from Vout1 and passes through a set of op-amps and two diodes (D1 and D2). In this case, the op-amp and diode configuration forms part of a precision rectifier. However, based on the polarity and arrangement of the diodes in the circuit, only the negative part of the signal from Vout1 is allowed to pass through to Vout2. Vout3 shows the signal after passing through the second op-amp, which functions to stabilize and restore the amplitude of the signal, producing a cleaner waveform. In a half-wave, only one part of the wave is released, while in a full-wave, both parts of the wave are reversed to the positive direction. This shows whether the rectification process is happening correctly. Observation at these three points is important to understand the entire signal processing process from amplification, rectification, to final stabilisation.

Table 4.1: Circuit Implementation in LTspice for Instrumentation amplifier with half-wave and full-wave rectifier

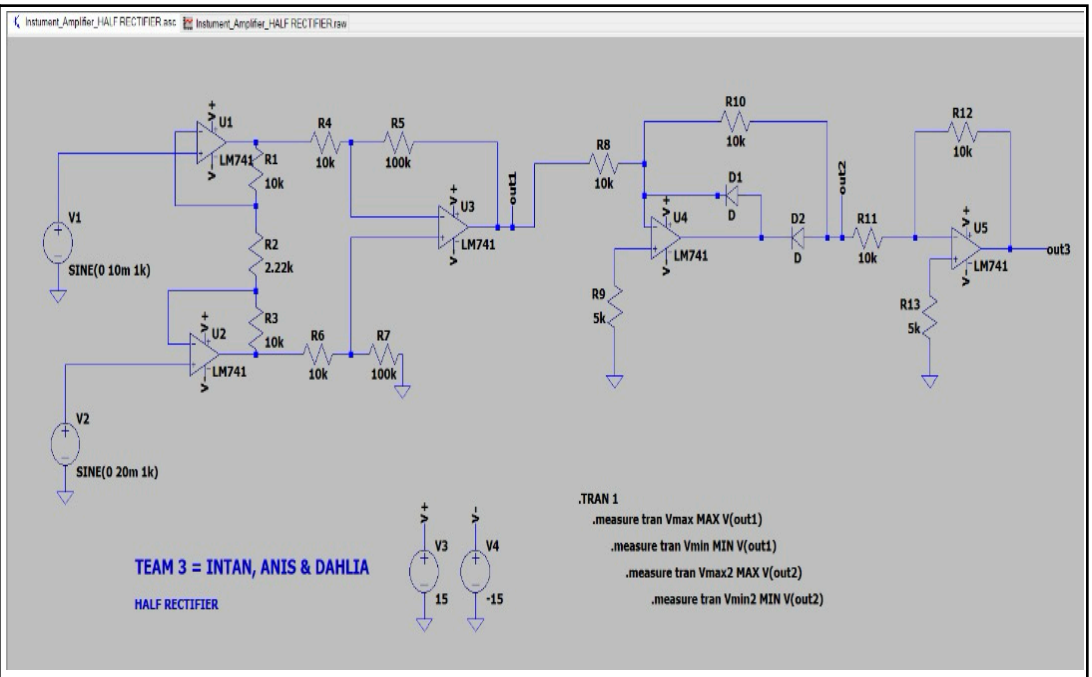
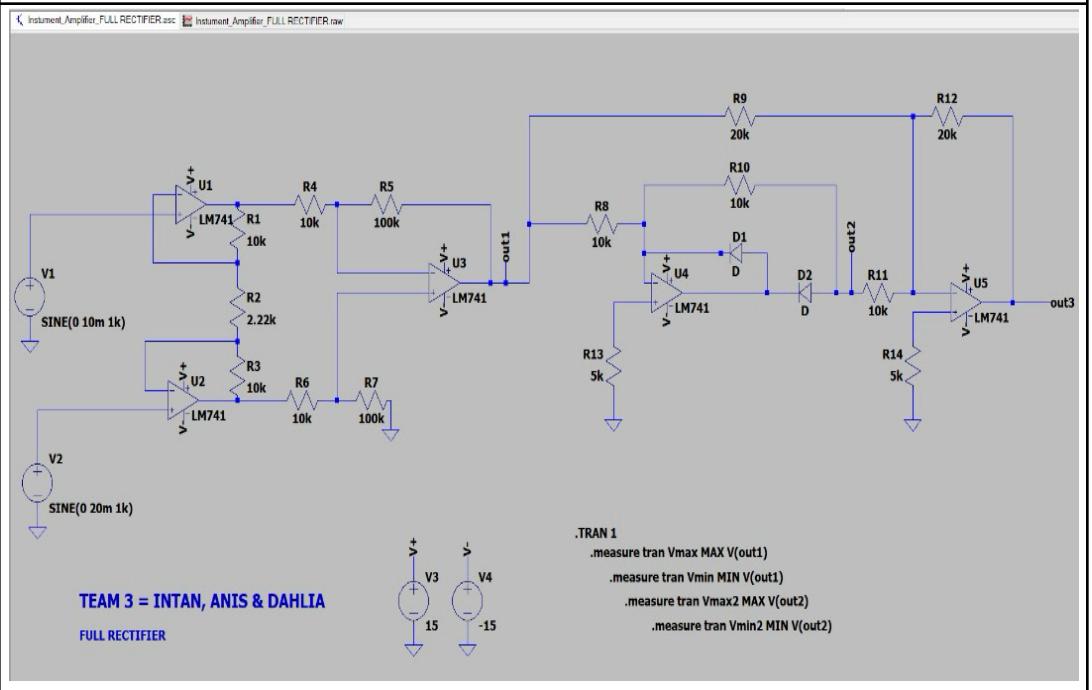
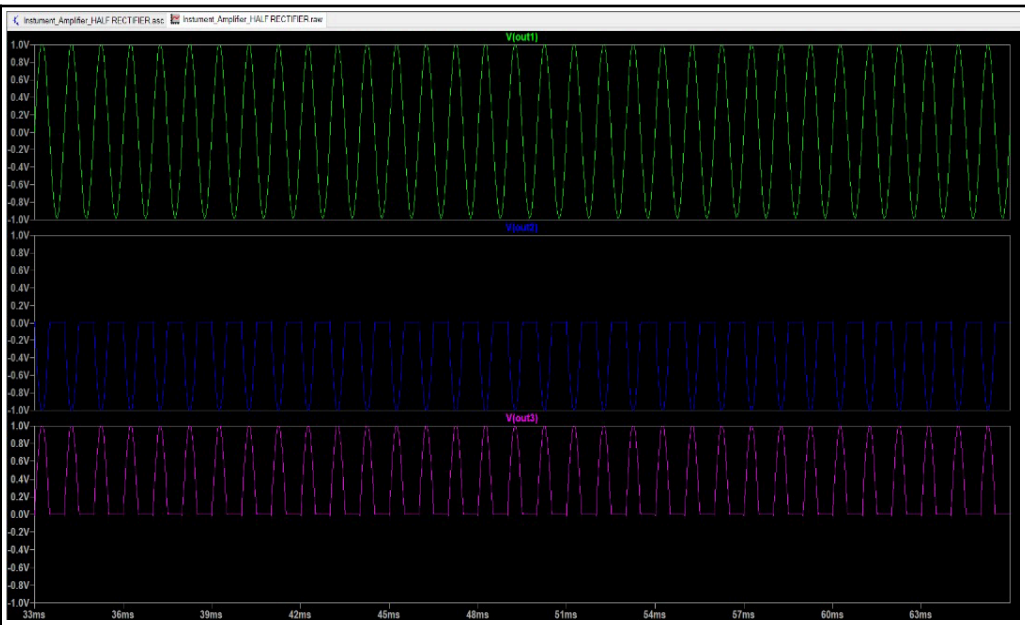
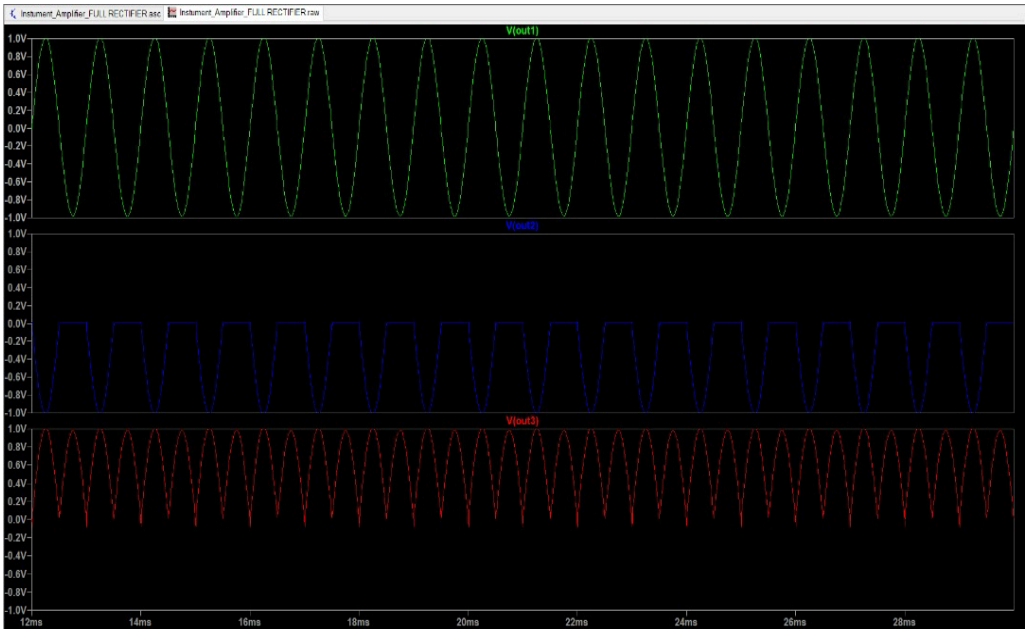
Half-wave	
Full-wave	

Table 4.2: Simulation Results in LTspice for Instrumentation amplifier with half-wave and full-wave rectifier

Half-wave	
Full-wave	

The circuit in Table 4.1 simulates an instrumentation amplifier with half-wave and full-wave rectifiers using LTspice. A 10 mV sinusoid input is used for the full-wave rectifier, while 20 mV is used for the half-wave rectifier to ensure sufficient signal amplitude after loss during filtering. The resistor values are chosen based on theoretical calculations to maintain the correct

gain and signal preparation. For example, the gain of an instrumentation amplifier is determined by the ratio of the feedback resistors, ensuring that the amplification matches the expected value. The rectifier diode is used to reduce the voltage drop, increasing accuracy. With the theoretical calculations above, the circuit connections are verified to be in line to ensure that the simulated output is also in line with the theoretical expectations, according to the measurements carried out.

Based on Table 4.2, the output of the half-wave rectifier shows that at Out1, there is a full sine wave resulting from the instrumentation amplifier. At Out2, only the negative part is visible while the positive part has been clipped to zero because the diode only allows the negative part to pass. Meanwhile, at Out3, the negative part is inverted to positive around 1V, while the positive part remains 0V, indicating the function of the second op-amp inverting and stabilising the signal. For the full-wave rectifier, Out1 still shows a full sine wave, but at Out2, only the negative part is maintained at -1 V while the positive part is cut off, indicating that the rectification process is not yet complete. However, at Out3, both the positive and negative parts have been inverted to a fully positive signal around 1V, proving that the circuit functions as a full-wave precision rectifier with the help of the second op-amp to invert and stabilise the entire wave to the positive side. This result is in accordance with the theoretical prediction, validating the circuit design. The small differences may be due to non-ideal diode characteristics or parasitic effects in LTspice. Overall, the simulation achieves the expected output for both configurations.

In the analysis of the results, the comparison between the simulation results and the theoretical calculations shows that the waveforms obtained for the half-wave and full-wave rectifiers are in line with the theoretical expectations. The waveform at Vout3 for the full-wave shows a more stable and complete signal, suitable for digital measurement or processing applications. However, the challenges faced include the accuracy of the simulation, which depends on the component model, and the difficulty of obtaining a perfect signal cut-off without small distortions at Vout2.

5. METHODOLOGY: BUTTERWORTH HIGH-PASS FILTER

One kind of filter design used in signal processing is the Butterworth filter. It is intended to provide a frequency response in the pass band that is as flat as is theoretically feasible. The two equations were solved using Butterworth, and four pole filters and demonstrated how vacuum tube amplifiers may be used to isolate the latter. High-pass filter is the complement of the low-pass filter. The frequency ranges from zero to cut-off frequency, fo is the stop band, and the range from fo to infinity from fo to infinity is the passband. In addition, a butterworth filter is a type of signal processing filter designed to have a frequency response as flat as possible in the passband. Filters are specified by two parameters, the cut-off frequency and the filter order. In this case, we obtain the cut-off frequency of the filter is 1MHz and the filter order is 5. However, the butterworth filter can be realized using passive components and active components such as resistors, capacitors and operational amplifiers with Sallen-key topology.

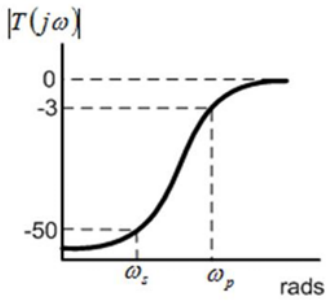
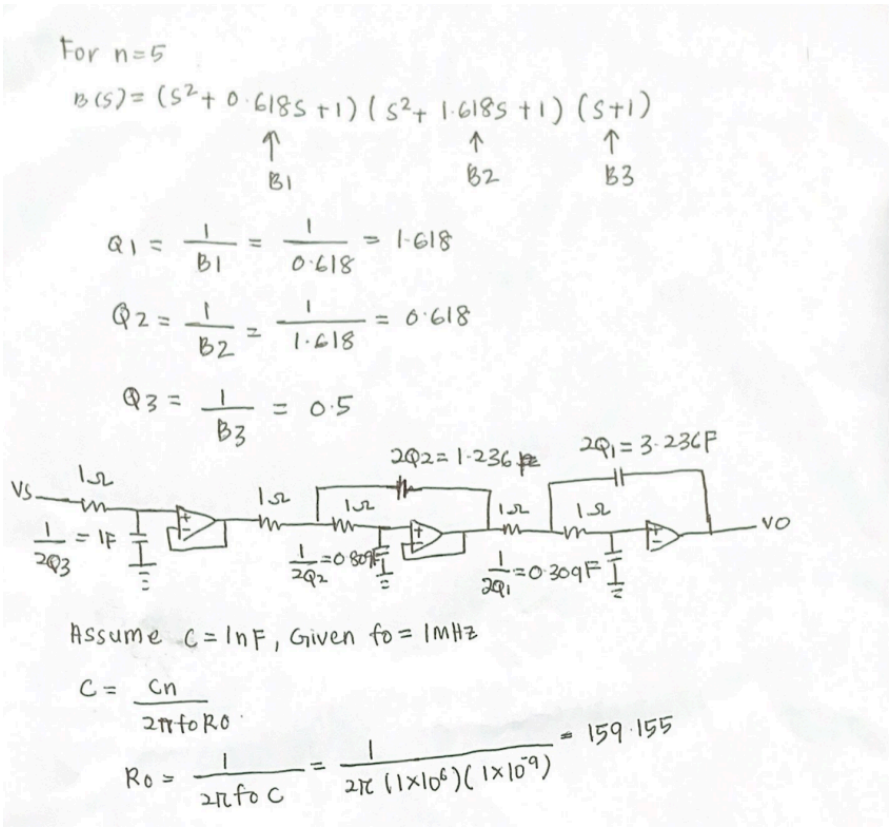


Figure 6.1 : Graph of High Pass Butterworth Filter

(c) Denominator Polynomial Factors $B(s) = B_1(s)\,B_2(s)\,B_3(s)\,B_4(s)\,B_5(s)$	
Order n	$B(s)$
1	$(s + 1)$
2	$(s^2 + 1.41421356s + 1)$
3	$(s^2 + s + 1)\quad (s + 1)$
4	$(s^2 + 0.76536686s + 1)\quad (s^2 + 1.84775907s + 1)$
5	$(s^2 + 0.61803399s + 1)\quad (s^2 + 1.61803399s + 1)\quad (s + 1)$

Figure 6.2: Butterworth Polynomials with respect to order of the filter, n

Since the question states filter order, n is 5. So, we obtain Polynomial Factors, $B(s) = (s^2 + 0.618s + 1)(s^2 + 1.618s + 1)(s + 1)$.



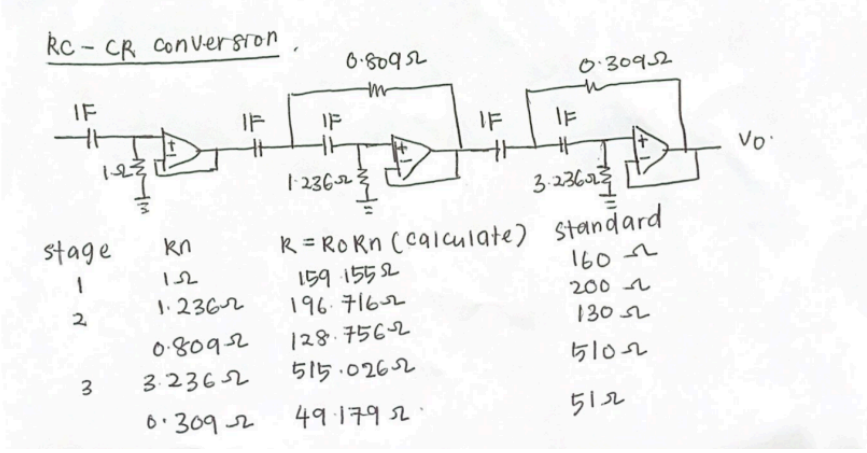


Figure 6.3: Mathematical theory Butterworth High-pass Filter

6. RESULTS AND DISCUSSION: BUTTERWORTH HIGH-PASS FILTER

This section will present the High-pass Butterworth filters by using LTSpice as the tools simulation and make a comparison between the mathematical theory and simulation. A High-pass filter must be designed according to specifications, to require being 5 order and has cutoff frequency 1MHz [8]. The filter was first implemented using an ideal operational amplifier, so as to observe typical characteristics free from limitations engendered by real components. The theoretical calculations were based on the maximally flat passband response of Butterworth filter, whereas the 5th order design was split into a first-order stage and two second-order stages, all in Sallen-Key configuration. In the ideal simulation, the filter demonstrated the expected behavior in terms of sharp roll-off at the cutoff frequency of 1 MHz, with slight variations from more explicit mathematical expressions.

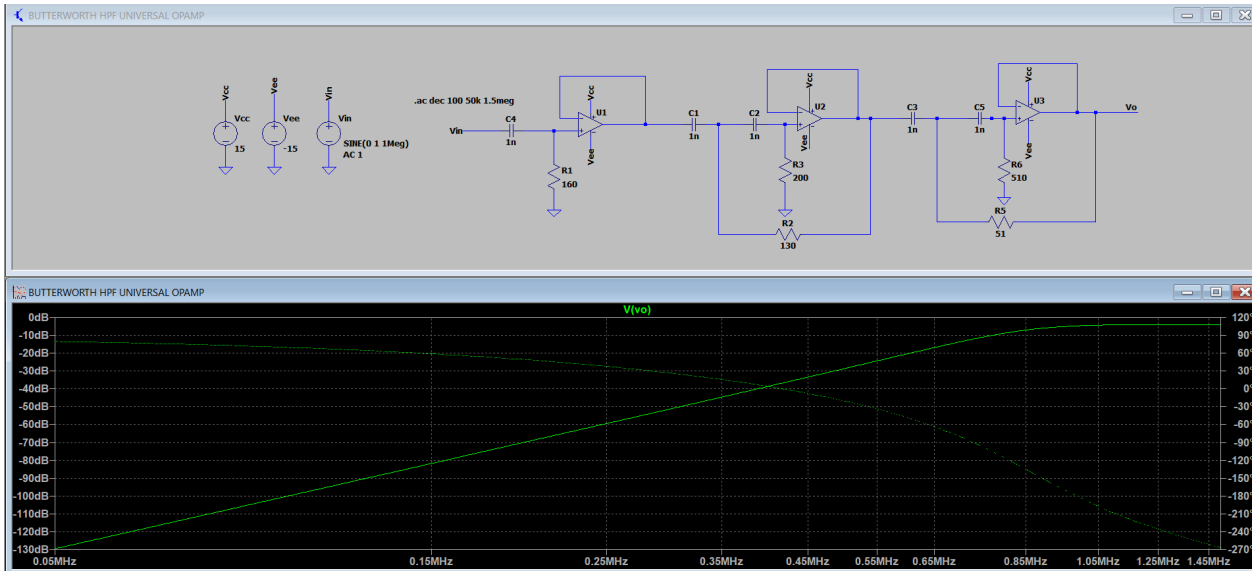


Figure 7.1: Butterworth High Pass-Filter using Universal Op-Amp

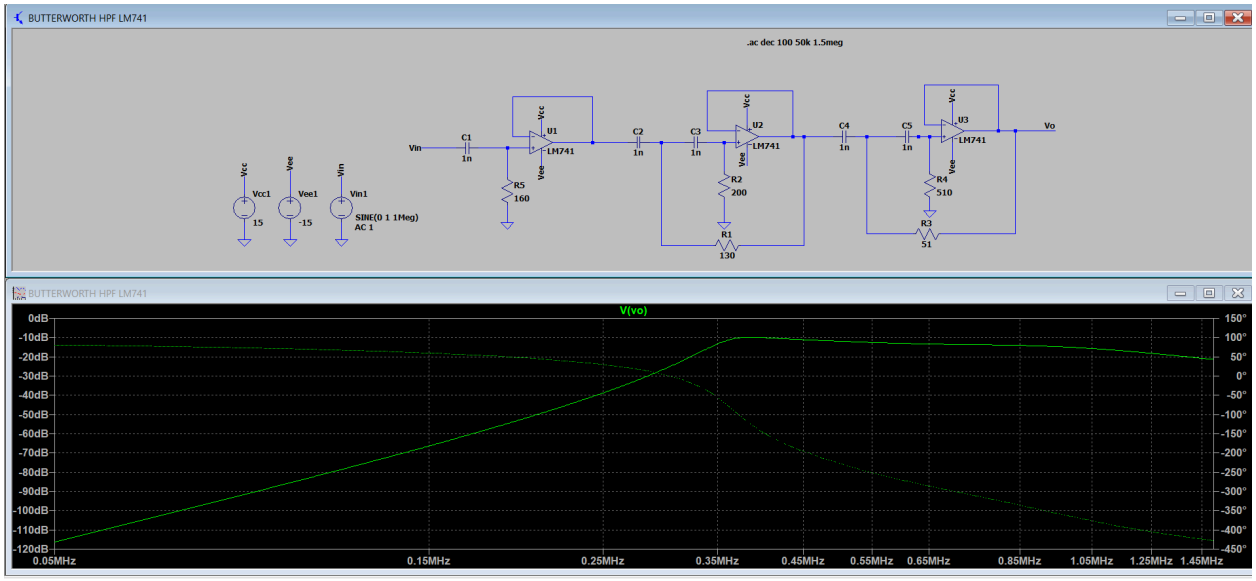


Figure 7.2: Butterworth High Pass-Filter using LM741

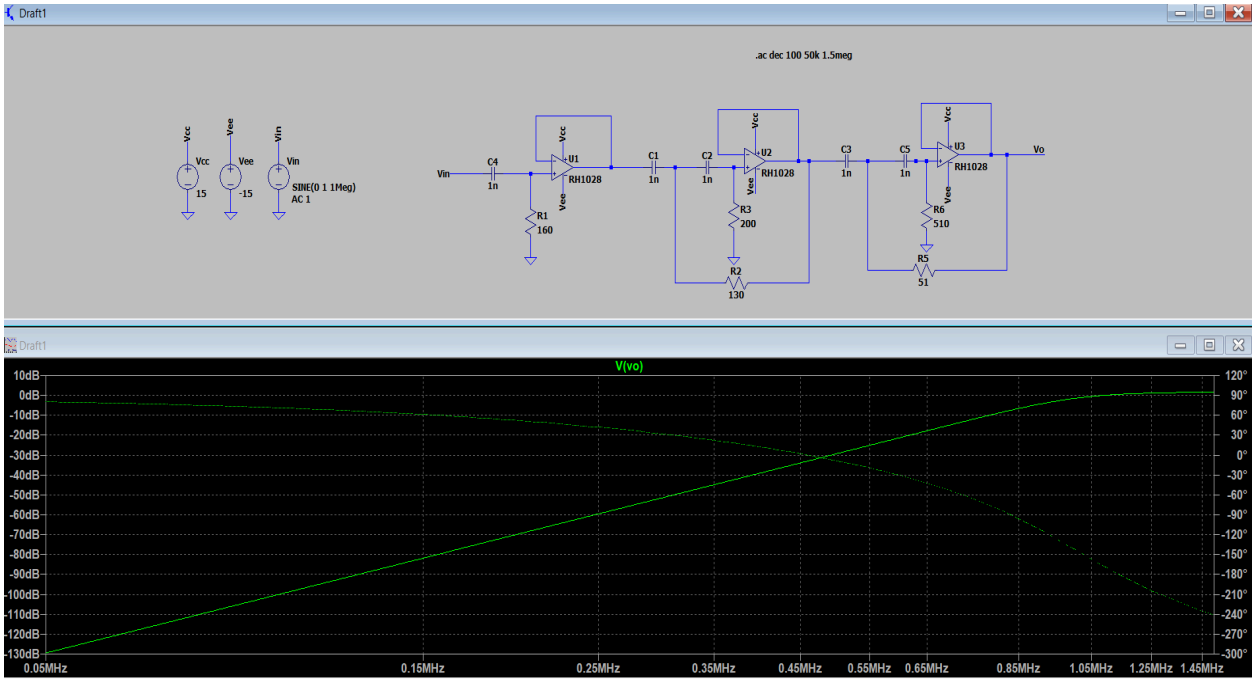


Figure 7.3: Butterworth High Pass-Filter RH1028

Parameter	Universal Op-Amp	LM741 Op-Amp	RH1028 Op-Amp
Gain-Bandwidth Product	∞	1MHz	10MHz
Input Noise Voltage	0 (Ideal)	18nV (noisy)	5nV (low noise)
Common-Mode Rejection (CMRR)	∞	70dB (moderate)	90 dB (excellent)
Slew Rate	∞	0.5V/μs	20V/μs

Ripple and distortion occur in a 1MHz high-pass Butterworth filter because of the LM741 op-amp's gain-bandwidth product (GBW) limit of 1MHz. Due to the filter's cutoff frequency matching the op-amp's GBW, the LM741 is unable to achieve flat gain over the critical transition region. This problem is further exacerbated by the op-amp's low slew rate of 0.5V/μs, which creates high-frequency phase distortion and signal degradation. Along with the rotary compensation structure type of the op-amp, which compromises high frequency performance for stability at lower frequencies, this greatly increases the amount of ripple introduced around the cutoff frequency, moving further away from a maximally flat response. To improve stability and frequency response, we used an op-amp such as the RH1028 however, would be better suited since it has a 10MHz GBW and faster 20V/μs slew rate, thus not introducing any ripple at 1MHz. This is a clear example of why component selection is important in active filter design. The LM741 seems ideal for low-frequency applications under 100kHz, but becomes difficult to use around its upper frequency limit.

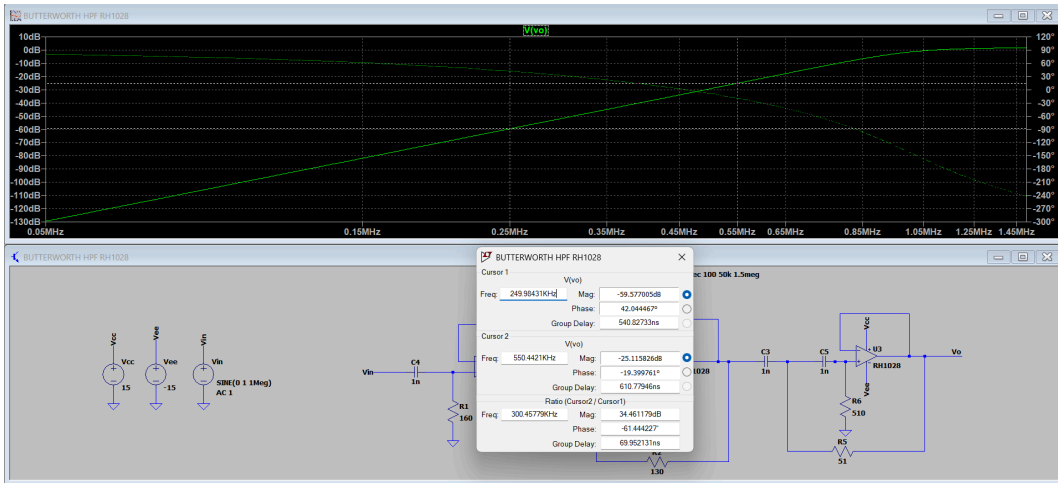


Figure 7.4: Fp and αp of RH1028 Op-Amp

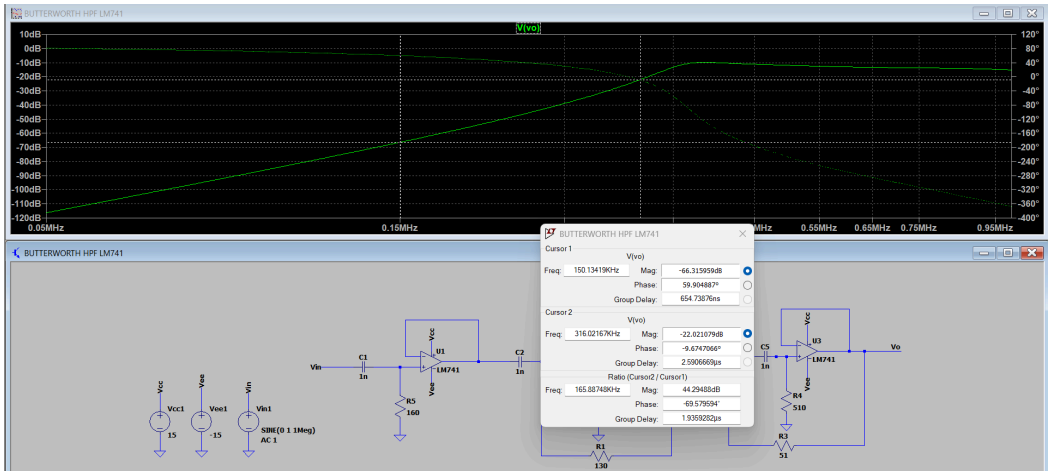


Figure 7.5:Fp and αp of LM741 Op-Amp

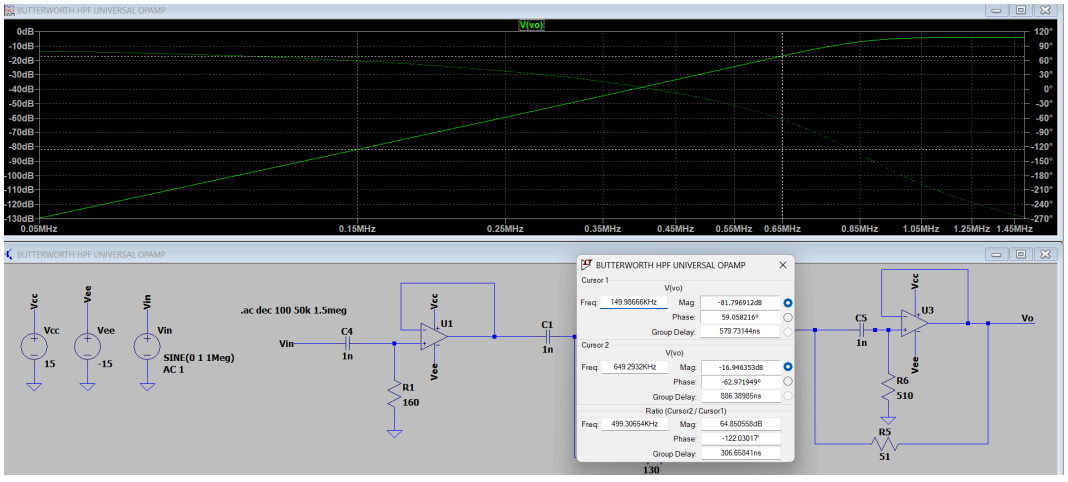


Figure 7.6: Fp and αp of Universal Op-Amp

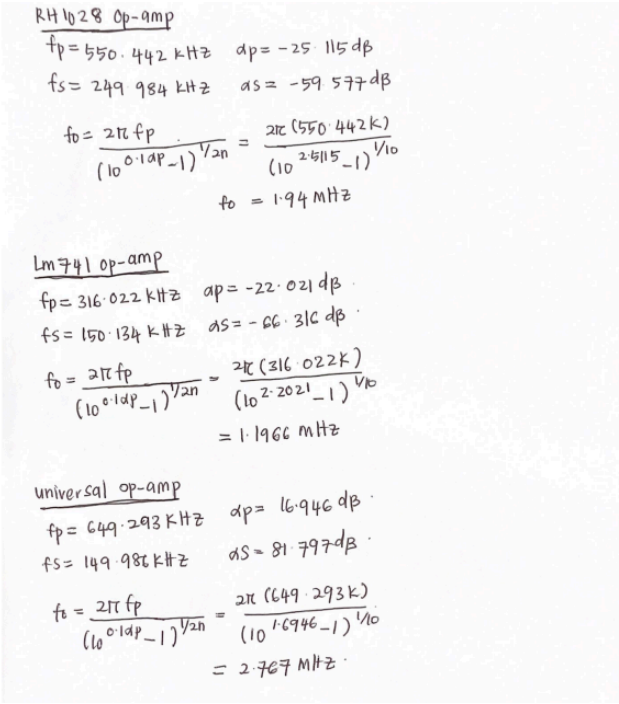


Figure 7.7: Calculation cutoff frequency RH1028, LM741 and Universal Op-Amp

In the simulations, one intriguing result was the performance of the LM741 op-amp which surpassed expectations with a cutoff frequency of 1.1966MHz, closely approaching the desired target of 1MHz despite having a gain-bandwidth limit of 1MHz. Yet, this figure is not as promising as it sounds, it stems from the LM741 not maintaining adequate gain at high frequencies, leading to early signal attenuation instead of genuine filter action. On the other hand, the RH1028 Op-Amp's measured cutoff of 1.94MHz and Universal Op-Amp's 2.767MHz while further away from the target, seem to be more accurate suggesting actual capabilities of the components based on their measurements. These concerning deviations indicate that there are some design aspects that should be adjusted or worked on further ‘fine-tuning’. The LM741’s “closest” result sacrifices excessive phase shift and distortion issues in the passband making it impossible to use. The higher cutoff from RH1028 suggests possible wrong calculations with the values of components, incorrect evaluation of implemented circuitry, and parasitic component effects. The extreme deviation from Universal Op-Amp probably came from working with the denormalization or inherent limitations of the circuit topology used for the calculations.

7. CONCLUSION

In conclusion, this mini project successfully demonstrated the design and simulation of a two-stage analog signal conditioning system using LTspice. The first task, which involved a 3-op-amp instrumentation amplifier combined with a precision rectifier, managed to amplify a weak 10 mV, 1 kHz sine wave and convert it into a usable DC signal. The second task focused on a 5th-order Butterworth high-pass filter that effectively removed low-frequency noise, achieving a clear and sharp frequency response. The circuit performed well during simulations, especially when using the RH1028 op-amp, which showed better accuracy and waveform clarity compared to the LM741. This system is suitable for applications such as sensor signal processing, biomedical instrumentation, and any system that requires clean amplification of small analog signals. For future work, hardware implementation could be explored to compare with simulation results, and alternative op-amps or filter configurations could be tested to further improve performance and reliability.

8. REFERENCES

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9. APPENDICES

LM741 Operational Amplifier

1 Features

- Overload Protection on the Input and Output
- No Latch-Up When the Common-Mode Range is Exceeded

2 Applications

- Comparators
- Multivibrators
- DC Amplifiers
- Summing Amplifiers
- Integrator or Differentiators
- Active Filters

3 Description

The LM741 series are general-purpose operational amplifiers which feature improved performance over industry standards like the LM709. They are direct, plug-in replacements for the 709C, LM201, MC1439, and 748 in most applications.

The amplifiers offer many features which make their application nearly foolproof: overload protection on the input and output, no latch-up when the common-mode range is exceeded, as well as freedom from oscillations.

The LM741C is identical to the LM741 and LM741A except that the LM741C has their performance ensured over a 0°C to +70°C temperature range, instead of −55°C to +125°C.

Device Information(1)

PART NUMBER	PACKAGE	BODY SIZE (NOM)
LM741	TO-99 (8)	9.08 mm × 9.08 mm
	CDIP (8)	10.16 mm × 6.502 mm
	PDIP (8)	9.81 mm × 6.35 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Typical Application

